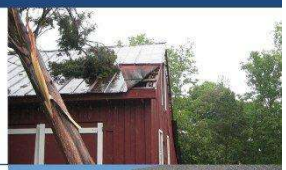
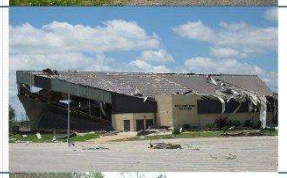
Tornado Dataset

Created By: Landry Ratliff







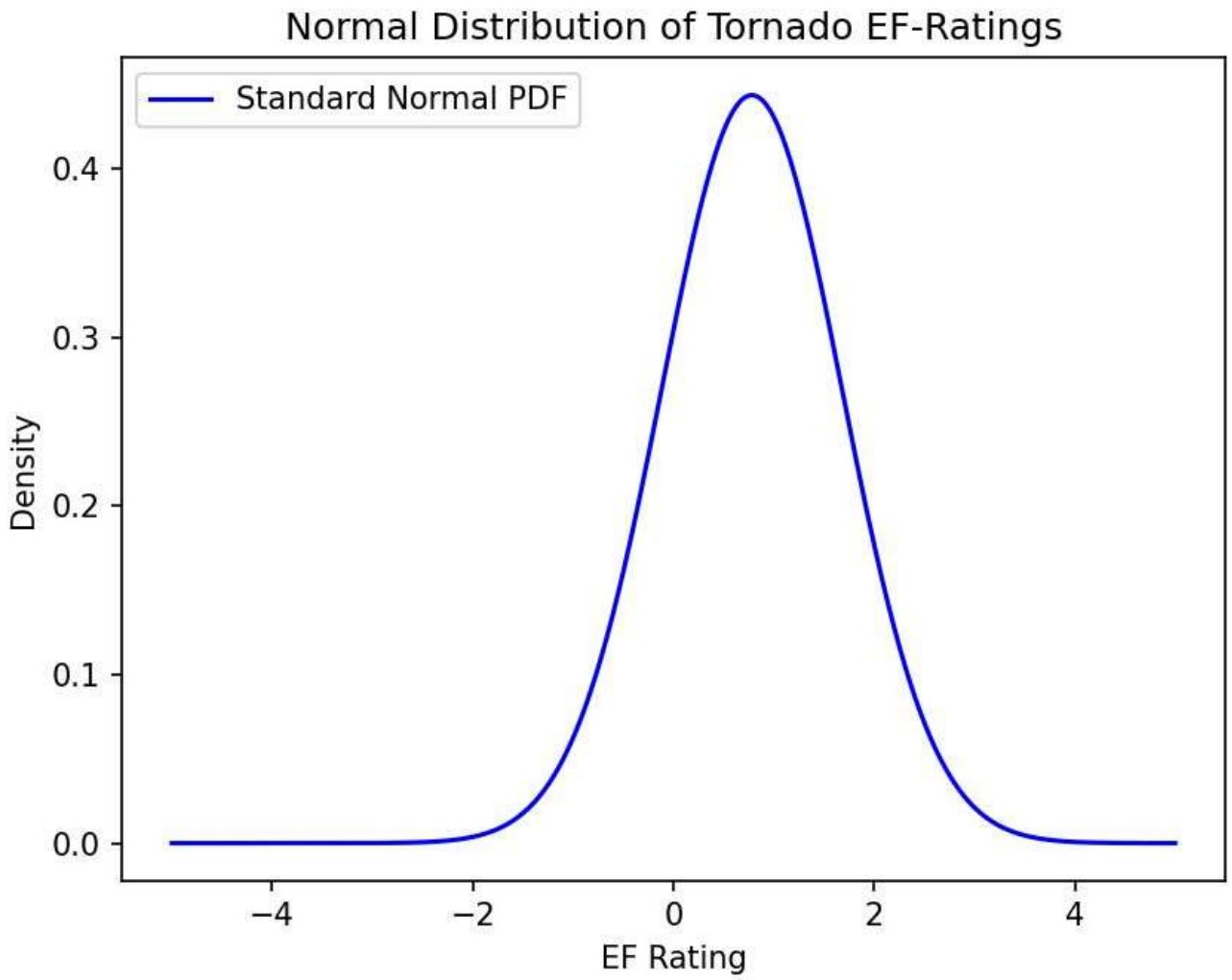
EF Rating	Wind Speeds	Expected Damage		
EF-0	65-85 mph	'Minor' damage: shingles blown off or parts of a roof peeled off, damage to gutters/siding, branches broken off trees, shallow rooted trees toppled.		
EF-1	86-110 mph	'Moderate' damage: more significant roof damage, windows broken, exterior doors damaged or lost, mobile homes overturned or badly damaged.		
EF-2	111-135 mph	'Considerable' damage: roofs torn off well constructed homes, homes shifted off their foundation, mobile homes completely destroyed, large trees snapped or uprooted, cars can be tossed.		
EF-3	136-165 mph	'Severe' damage: entire stories of well constructed homes destroyed, significant damage done to large buildings, homes with weak foundations can be blown away, trees begin to lose their bark.		
EF-4	166-200 mph	'Extreme' damage: Well constructed homes are leveled, cars are thrown significant distances, top story exterior walls of masonry buildings would likely collapse.		
EF-5	> 200 mph	'Massive/incredible' damage: Well constructed homes are swept away, steel-reinforced concrete structures are critically damaged, high-rise buildings sustain severe structural damage, trees are usually completely debarked, stripped of branches and snapped.		

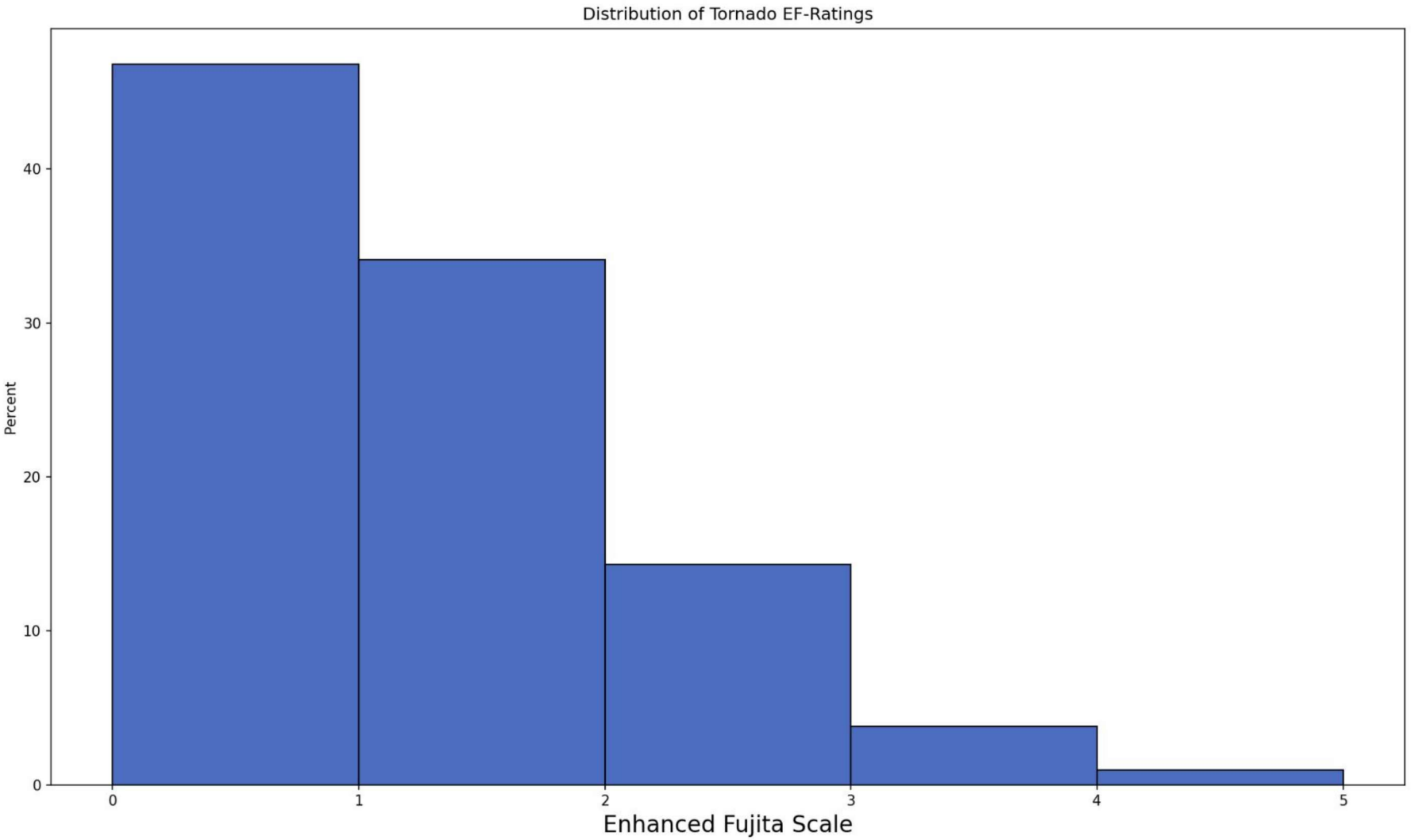
---Most Powerful Tornadoes---

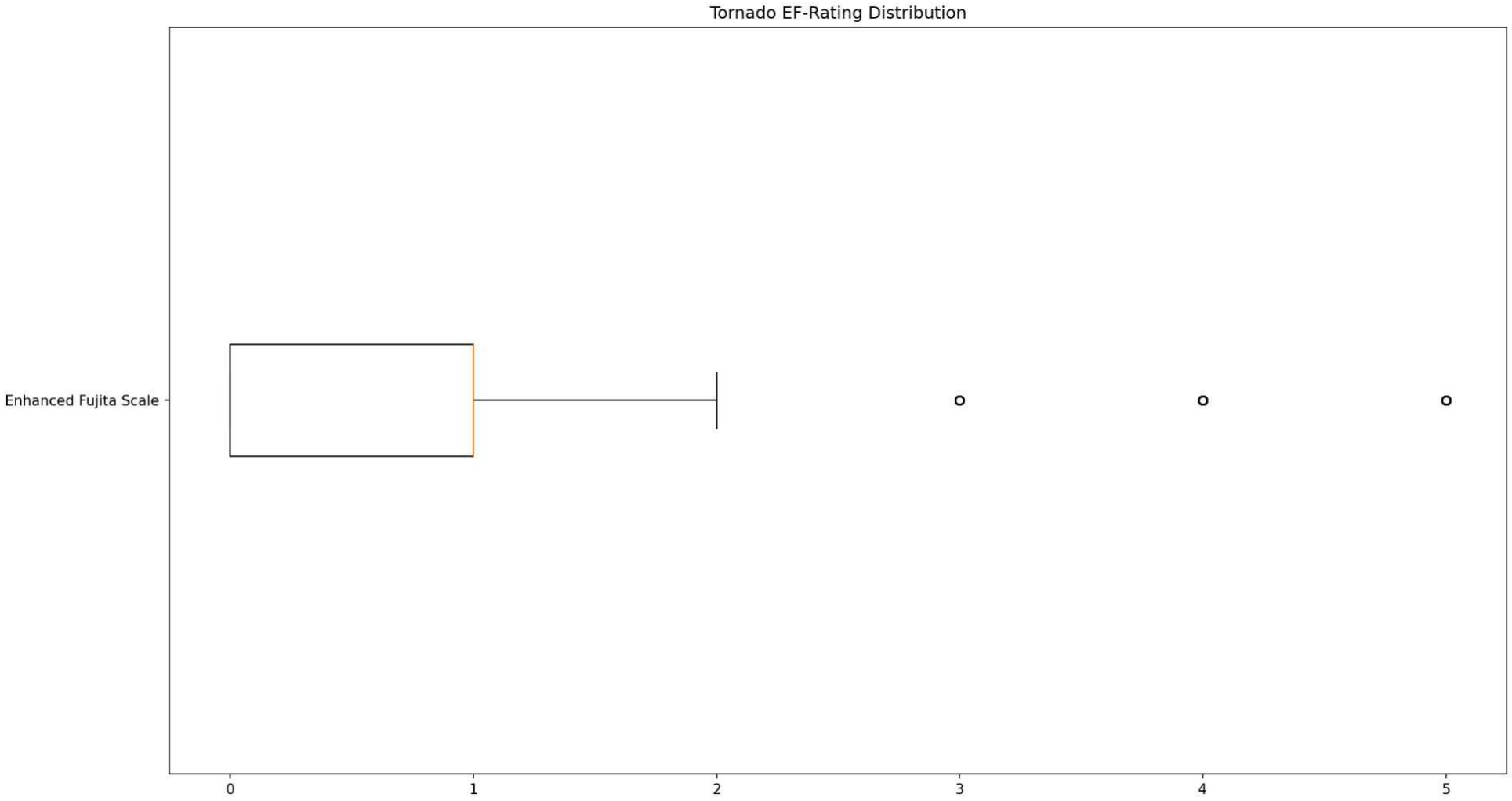
Date State	Date	Decimal Date	Decade	Year	Month	Day	State	Enhanced Fujita Scale	Wind Speed (mph)	Damage	Injuries	Fatalities
2013-05-20 Oklahoma	2013-05-20	2013.38	2010	2013	5	20	Oklahoma	5	>200 mph	Massive Damage	212	24
2011-05-24 Oklahoma	2011-05-24	2011.39	2010	2011	5	24	Oklahoma	5	>200 mph	Massive Damage	181	9
2011-05-22 Missouri	2011-05-22	2011.39	2010	2011	5	22	Missouri	5	>200 mph	Massive Damage	1150	158
2011-04-27 Alabama	2011-04-27	2011.32	2010	2011	4	27	Alabama	5	>200 mph	Massive Damage	0	25
2011-04-27 Alabama	2011-04-27	2011.32	2010	2011	4	27	Alabama	5	>200 mph	Massive Damage	145	72
2011-04-27 Mississippi	2011-04-27	2011.32	2010	2011	4	27	Mississippi	5	>200 mph	Massive Damage	6	3
2011-04-27 Mississippi	2011-04-27	2011.32	2010	2011	4	27	Mississippi	5	>200 mph	Massive Damage	137	23
2008-05-25 Iowa	2008-05-25	2008.4	2000	2008	5	25	Iowa	5	>200 mph	Massive Damage	70	9
2007-05-04 Kansas	2007-05-04	2007.34	2000	2007	5	4	Kansas	5	>200 mph	Massive Damage	63	11
1999-05-03 Oklahoma	1999-05-03	1999.34	1990	1999	5	3	Oklahoma	5	>200 mph	Massive Damage	583	36

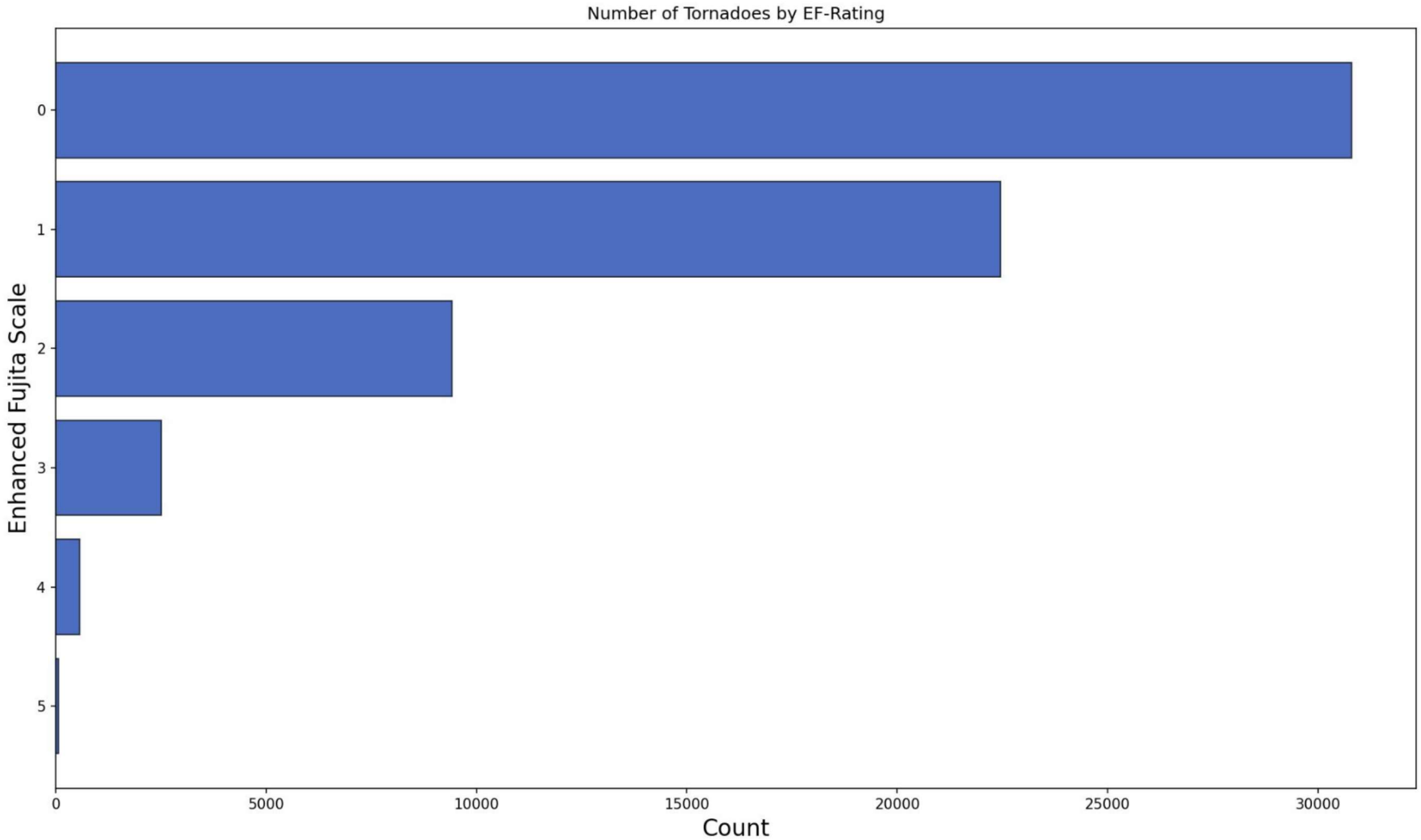
```
Summary Statistics:
Maximum EF Rating: 5
Average EF Rating: 0.78
Standard Deviation EF Rating: 0.9
Median EF Rating: 1.0
Minimum EF Rating: 0
Coefficient of Variation: 115.12%
```

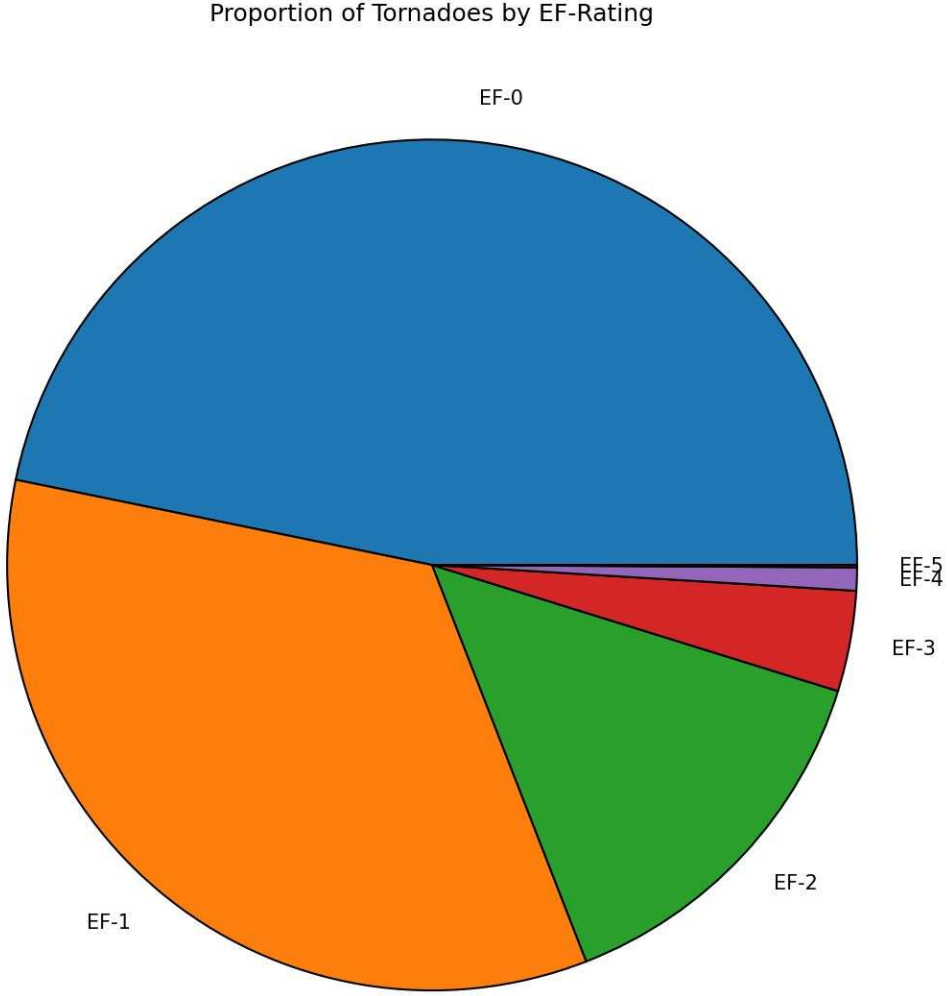
```
Average Number of Tornadoes per Year:
926.943661972
Average Number of Tornadoes per Month:
1:23.943661971830984
2:27.225352112676056
3:61.38028169014085
4:133.2112676056338
5:202.38028169014083
6:173.66197183098592
7:95.78873239436619
8:64.74647887323944
9:48.15492957746479
10:37.19718309859155
11:36.91549295774648
12:22.338028169014084
Average Number of Tornadoes per Day:
2.5382004705156387
Average Number of Deaths per Year:
84.6197183099
Average Number of Deaths per Day:
0.231709668711
```

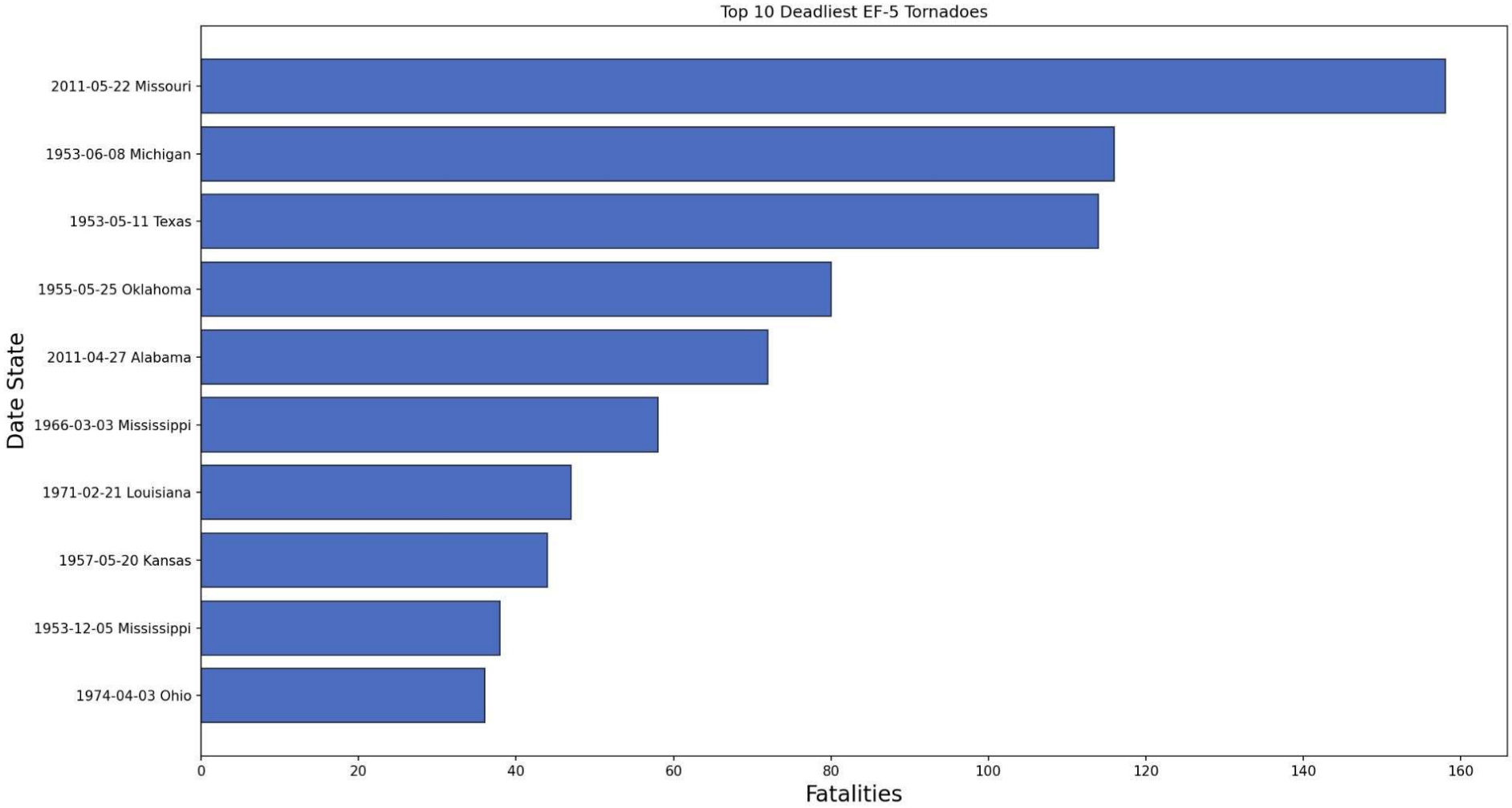

Normal Distribution of Tornado EF-Ratings
Standard Normal PDF
Density
0.0
0.1
0.2
0.3
0.4
−4
−2
0
2
4
EF Rating

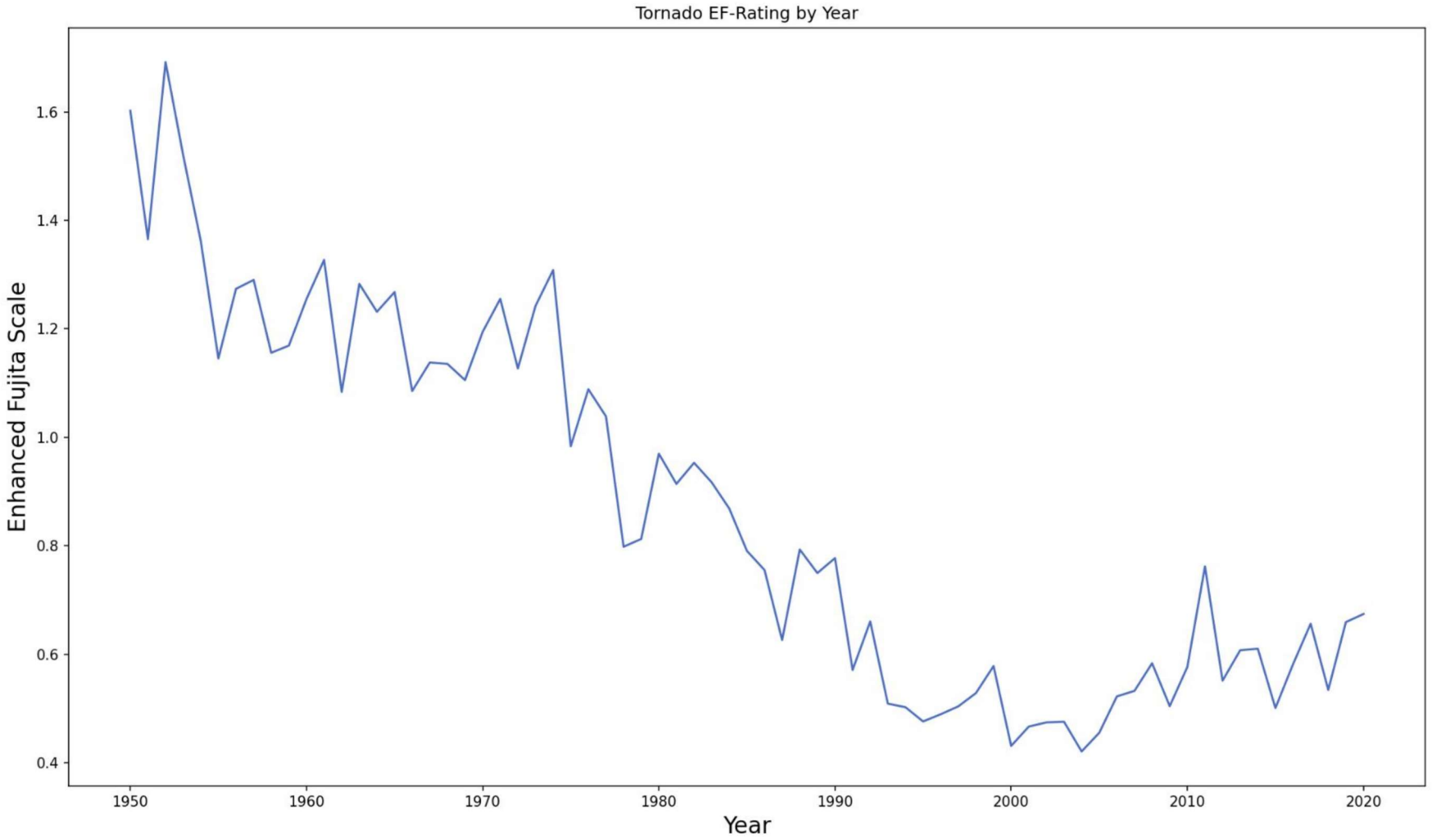

Distribution of Tornado EF-Ratings
Percent
0
10
20
30
40
0
1
2
3
4
5
Enhanced Fujita Scale

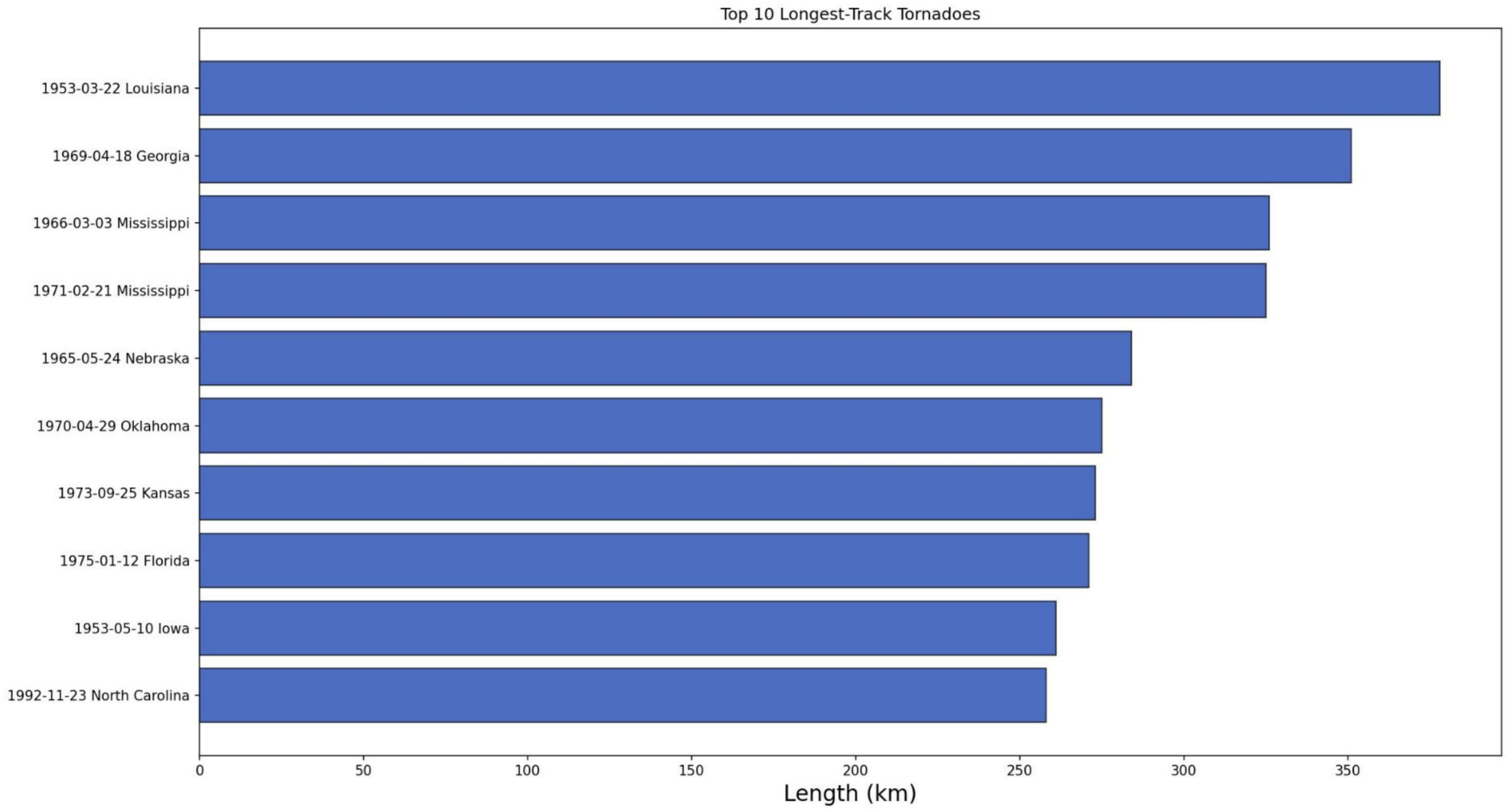

Tornado EF-Rating Distribution
Enhanced Fujita Scale
0
1
2
3
4
5

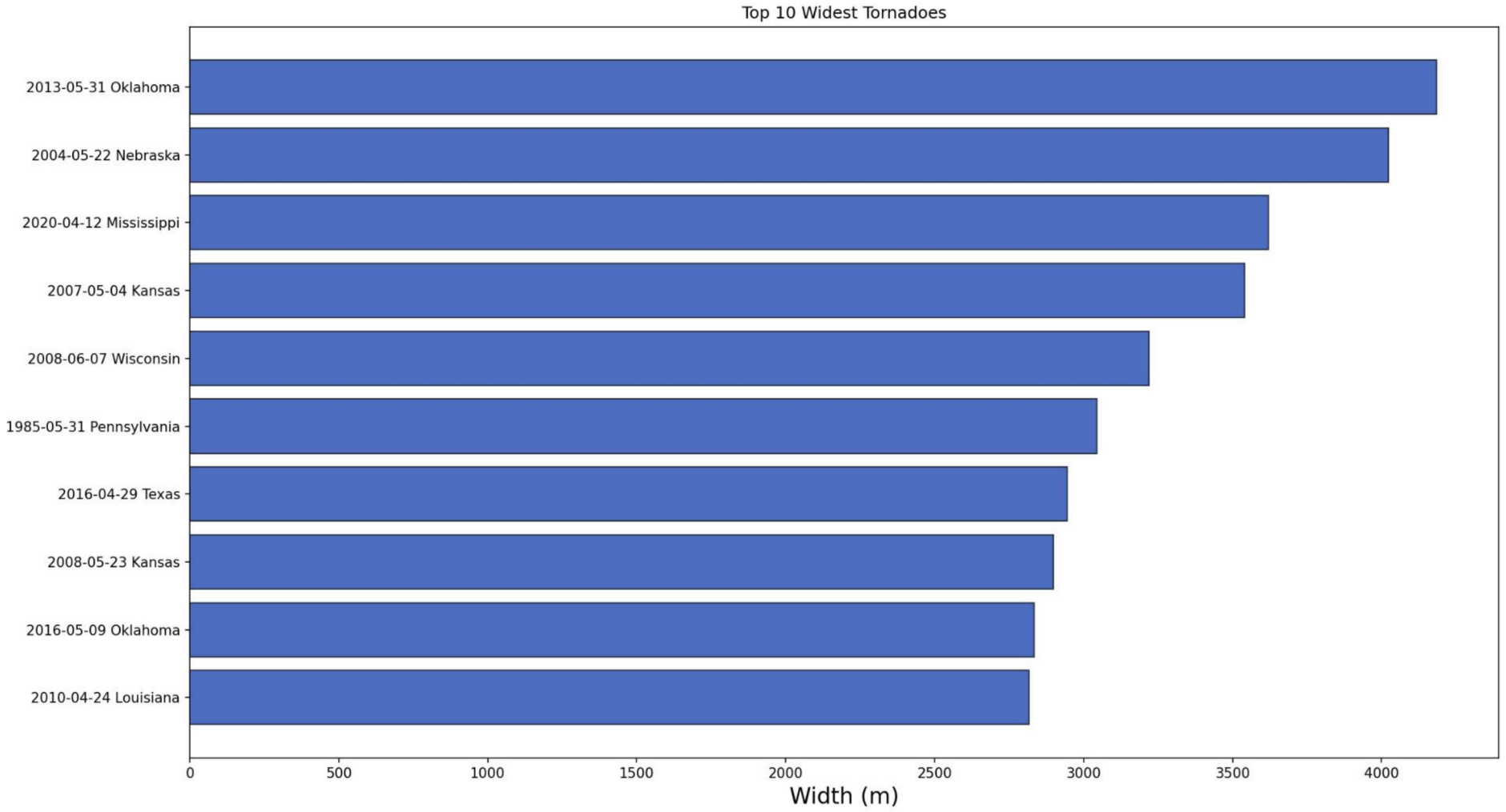

Number of Tornadoes by EF-Rating
Enhanced Fujita Scale
0
1
2
3
4
5
0
5000
10000
15000
20000
25000
30000
Count

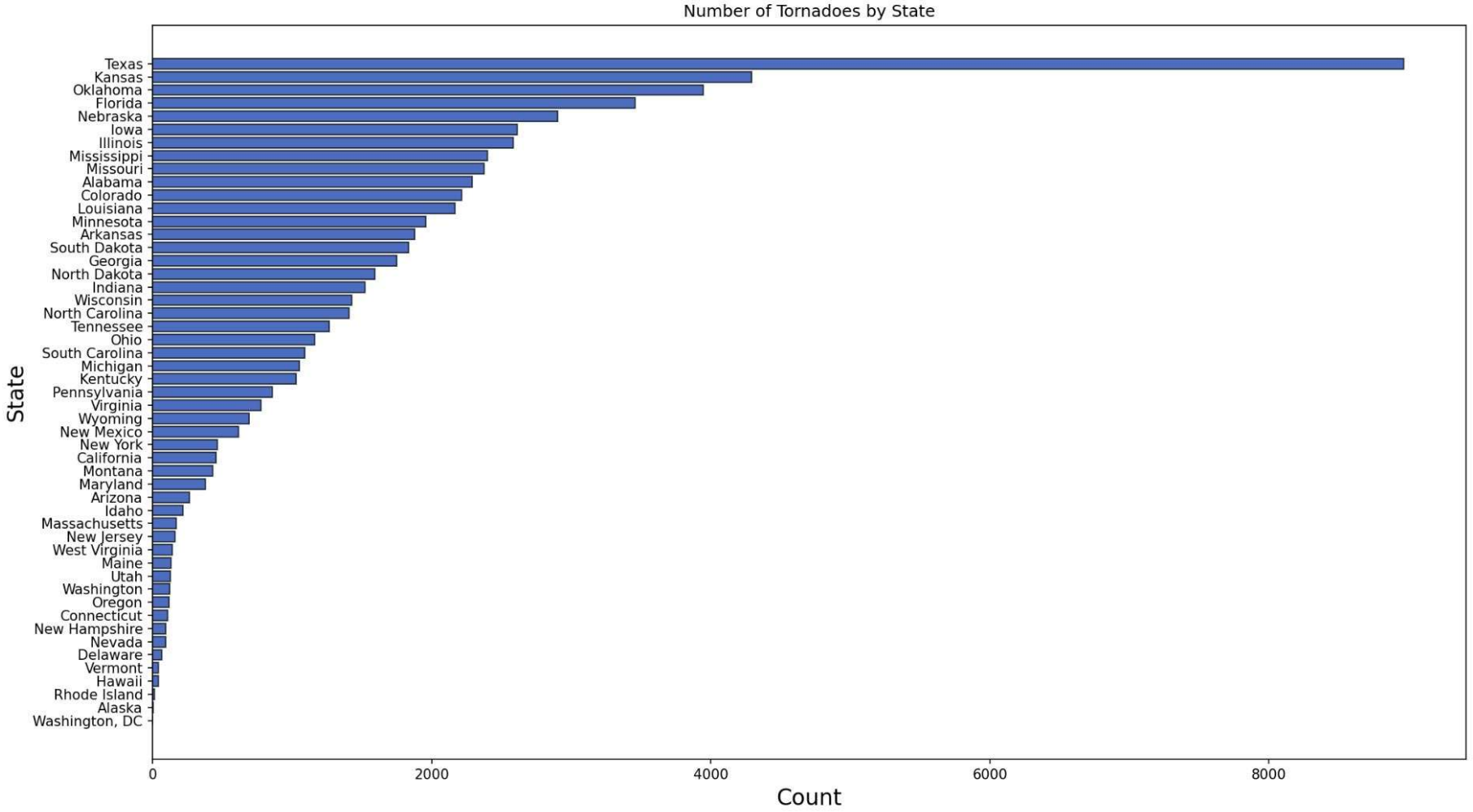

Proportion of Tornadoes by EF-Rating
EF-0
EF-1
EF-2
EF-3
EF-4
EF-5

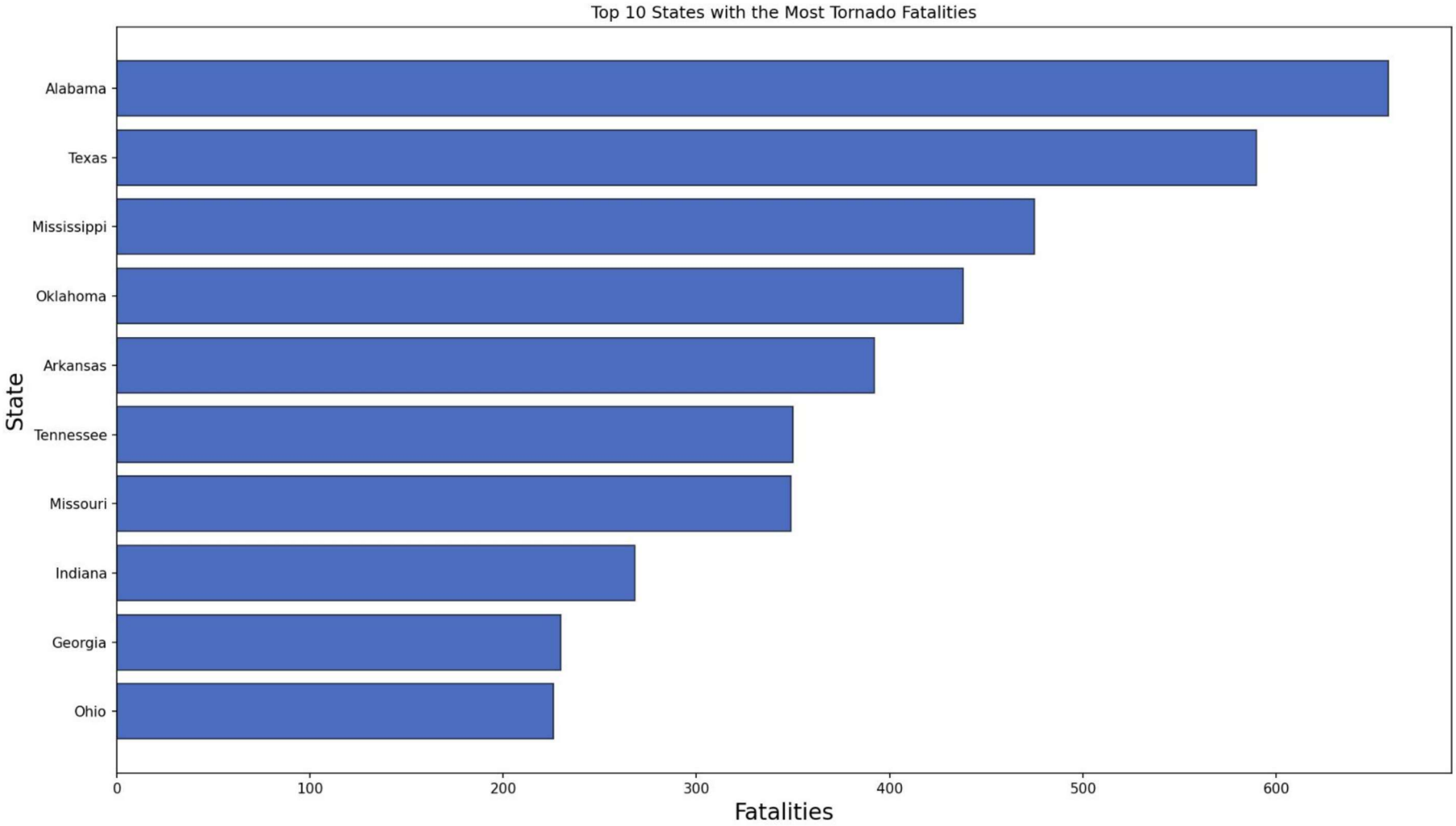

Top 10 Deadliest EF-5 Tornadoes
Date State
2011-05-22 Missouri
1953-06-08 Michigan
1953-05-11 Texas
1955-05-25 Oklahoma
2011-04-27 Alabama
1966-03-03 Mississippi
1971-02-21 Louisiana
1957-05-20 Kansas
1953-12-05 Mississippi
1974-04-03 Ohio
0
20
40
60
80
100
120
140
160
Fatalities

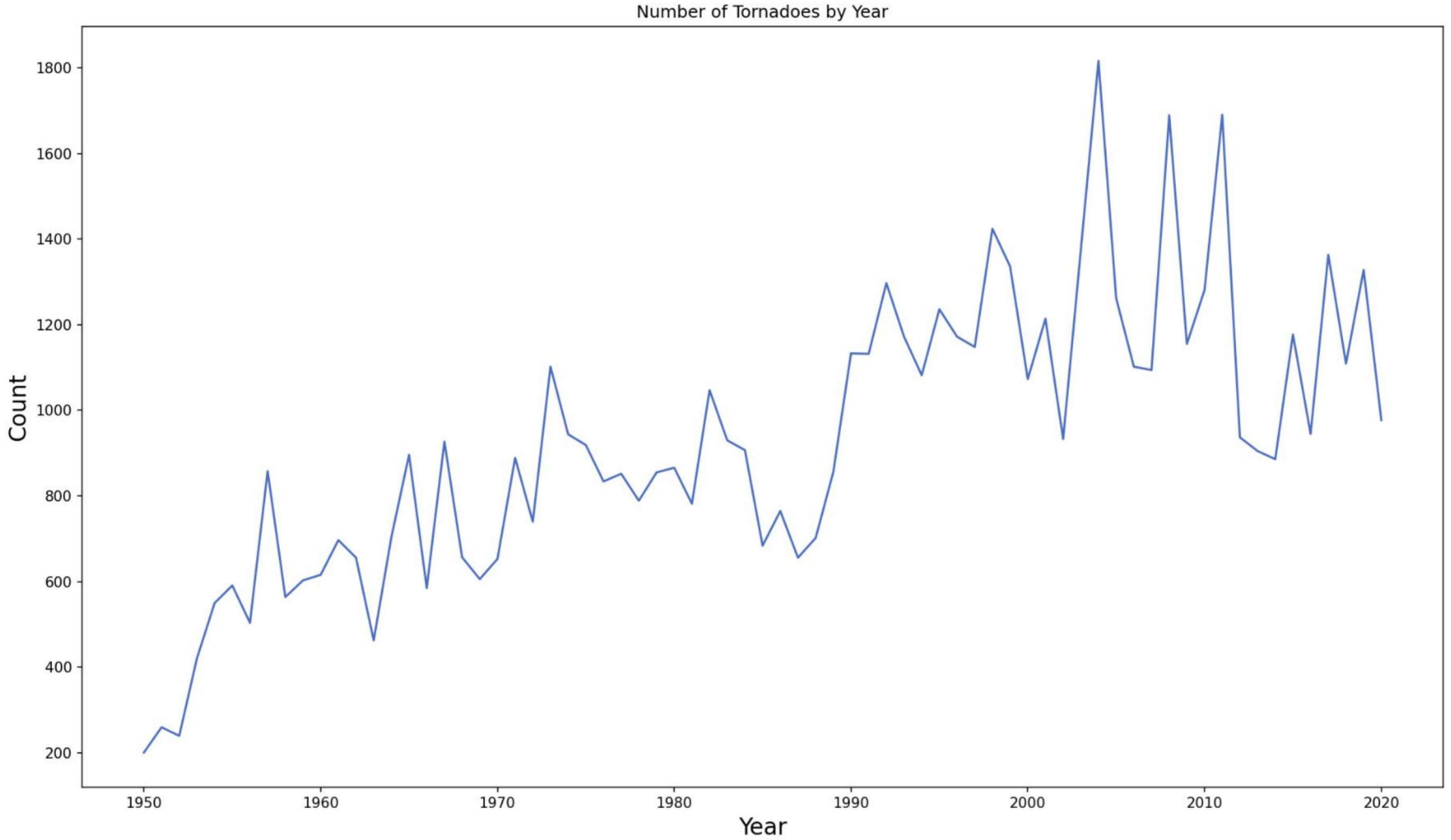

Tornado EF-Rating by Year
Enhanced Fujita Scale
1.6
1.4
1.2
1.0
0.8
0.6
0.4
1950
1960
1970
1980
1990
2000
2010
2020
Year

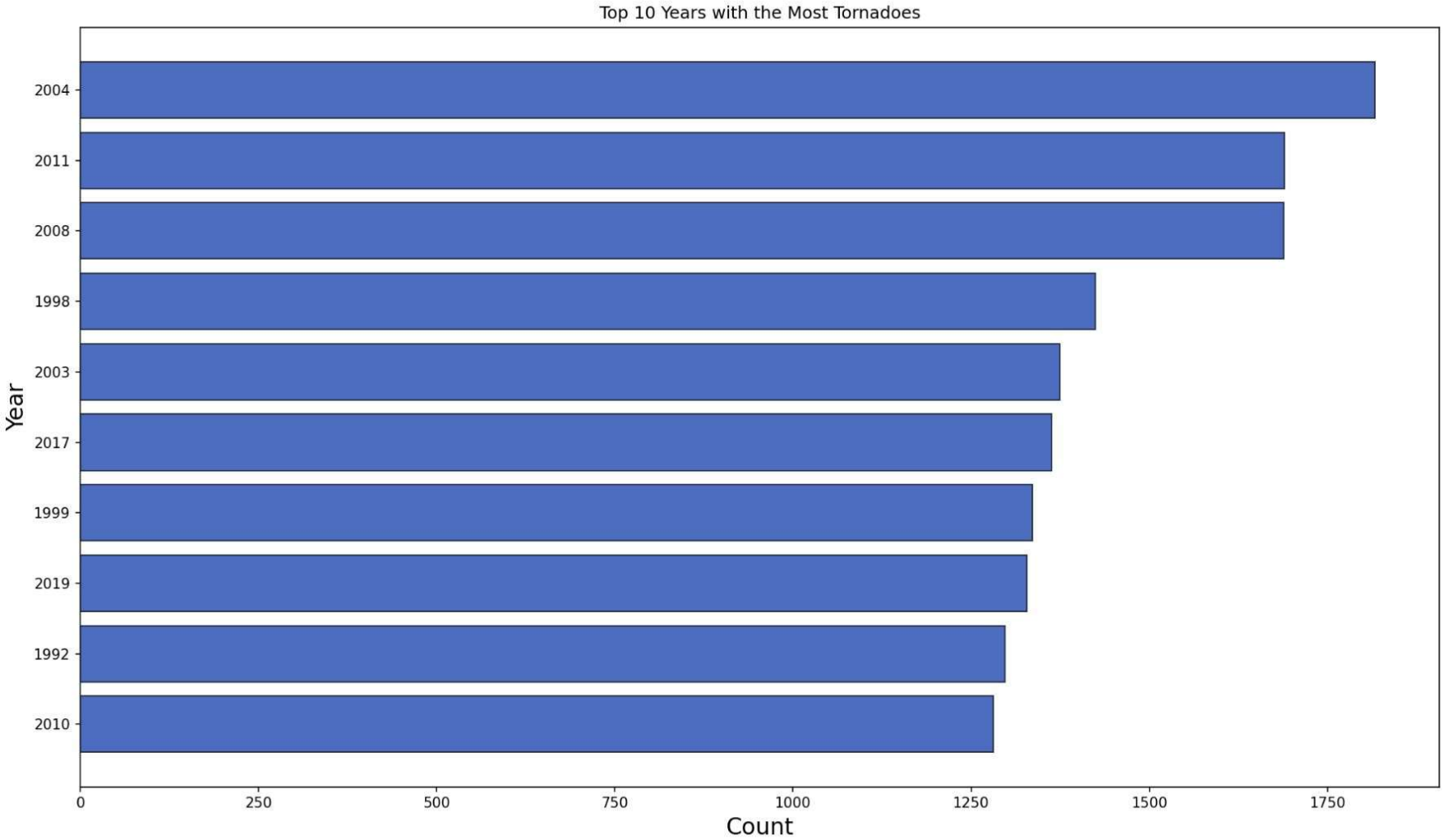

Top 10 Longest-Track Tornadoes
1953-03-22 Louisiana
1969-04-18 Georgia
1966-03-03 Mississippi
1971-02-21 Mississippi
1965-05-24 Nebraska
1970-04-29 Oklahoma
1973-09-25 Kansas
1975-01-12 Florida
1953-05-10 Iowa
1992-11-23 North Carolina
0
50
100
150
200
250
300
350
Length (km)

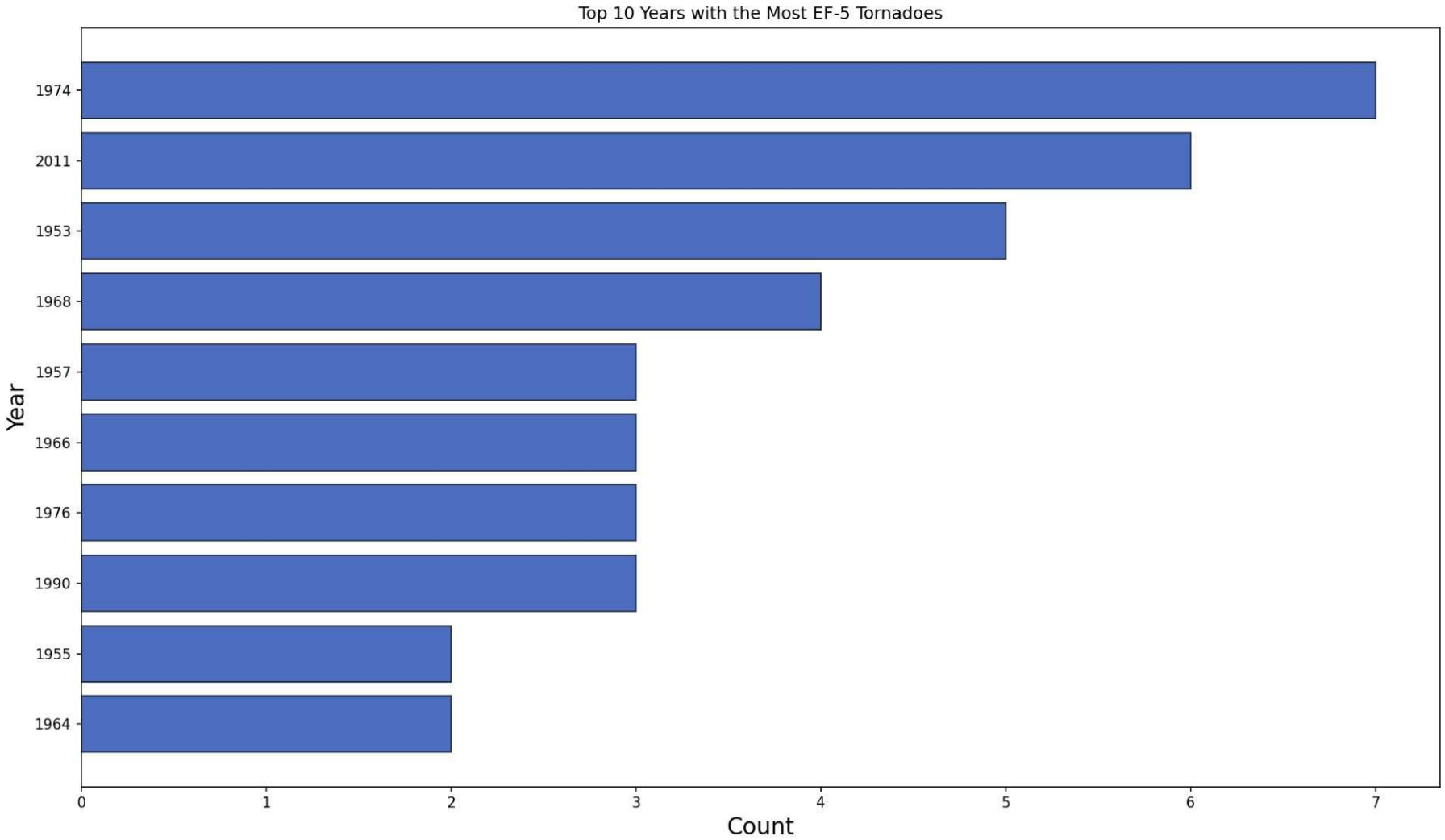

Top 10 Widest Tornadoes
2013-05-31 Oklahoma
2004-05-22 Nebraska
2020-04-12 Mississippi
2007-05-04 Kansas
2008-06-07 Wisconsin
1985-05-31 Pennsylvania
2016-04-29 Texas
2008-05-23 Kansas
2016-05-09 Oklahoma
2010-04-24 Louisiana
0
500
1000
1500
2000
2500
3000
3500
4000
Width (m)

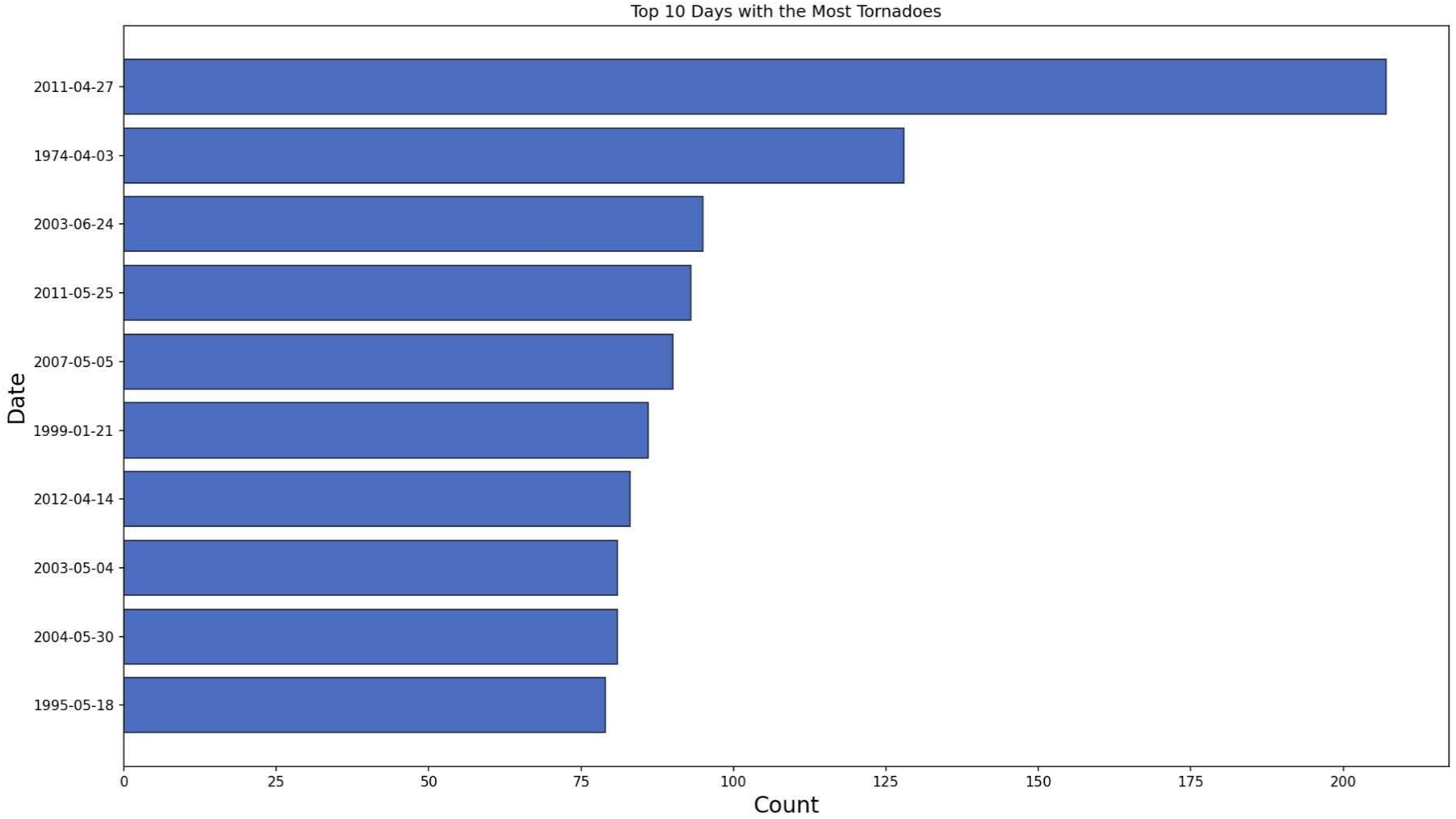

Number of Tornadoes by State
State
Texas
Kansas
Oklahoma
Florida
Nebraska
Iowa
Illinois
Mississippi
Missouri
Alabama
Colorado
Louisiana
Minnesota
Arkansas
South Dakota
Georgia
North Dakota
Indiana
Wisconsin
North Carolina
Tennessee
Ohio
South Carolina
Michigan
Kentucky
Pennsylvania
Virginia
Wyoming
New Mexico
New York
California
Montana
Maryland
Arizona
Idaho
Massachusetts
New Jersey
West Virginia
Maine
Utah
Washington
Oregon
Connecticut
New Hampshire
Nevada
Delaware
Vermont
Hawaii
Rhode Island
Alaska
Washington, DC
0
2000
4000
6000
8000
Count

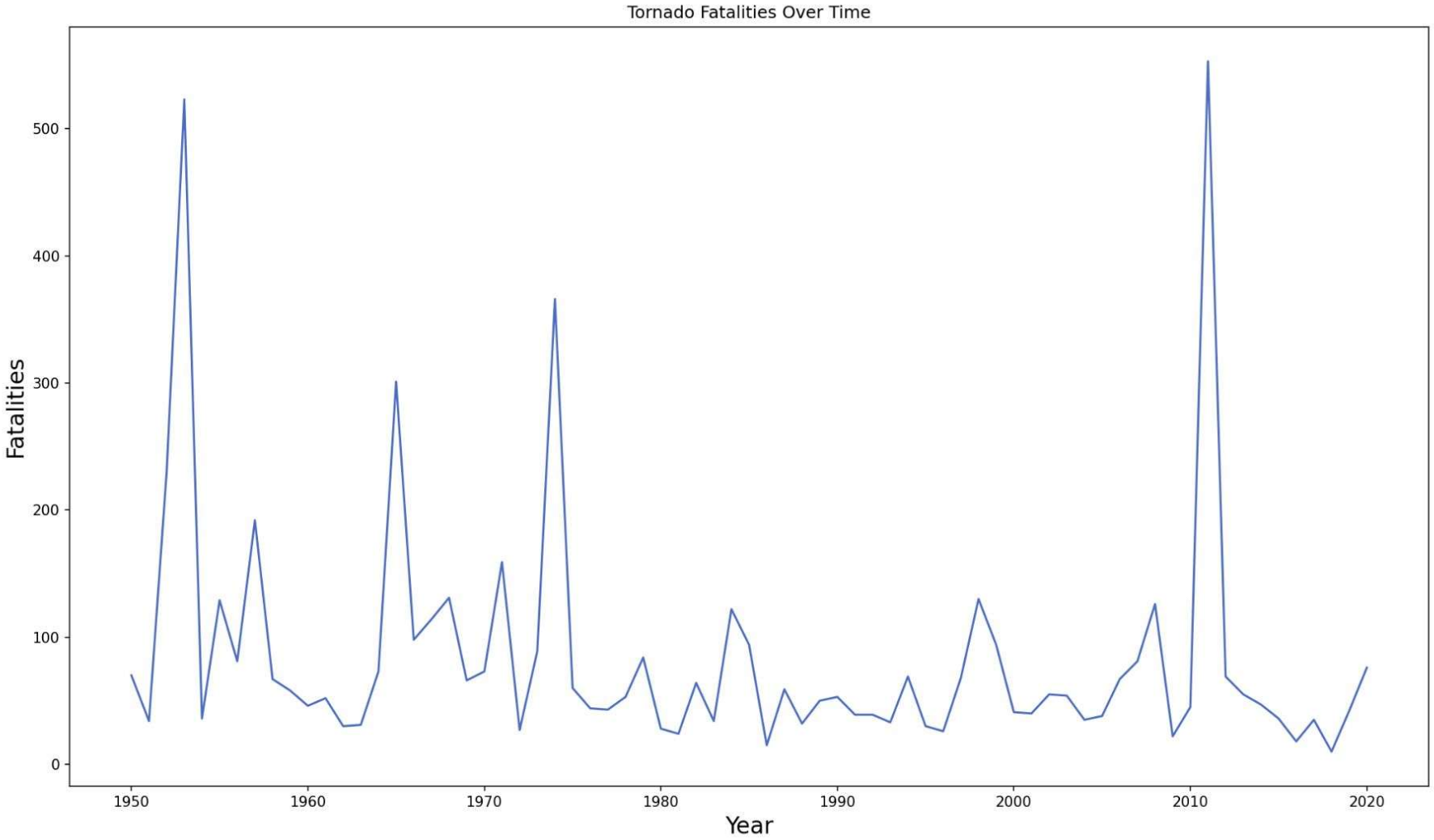

Top 10 States with the Most Tornado Fatalities
State
Alabama
Texas
Mississippi
Oklahoma
Arkansas
Tennessee
Missouri
Indiana
Georgia
Ohio
0
100
200
300
400
500
600
Fatalities

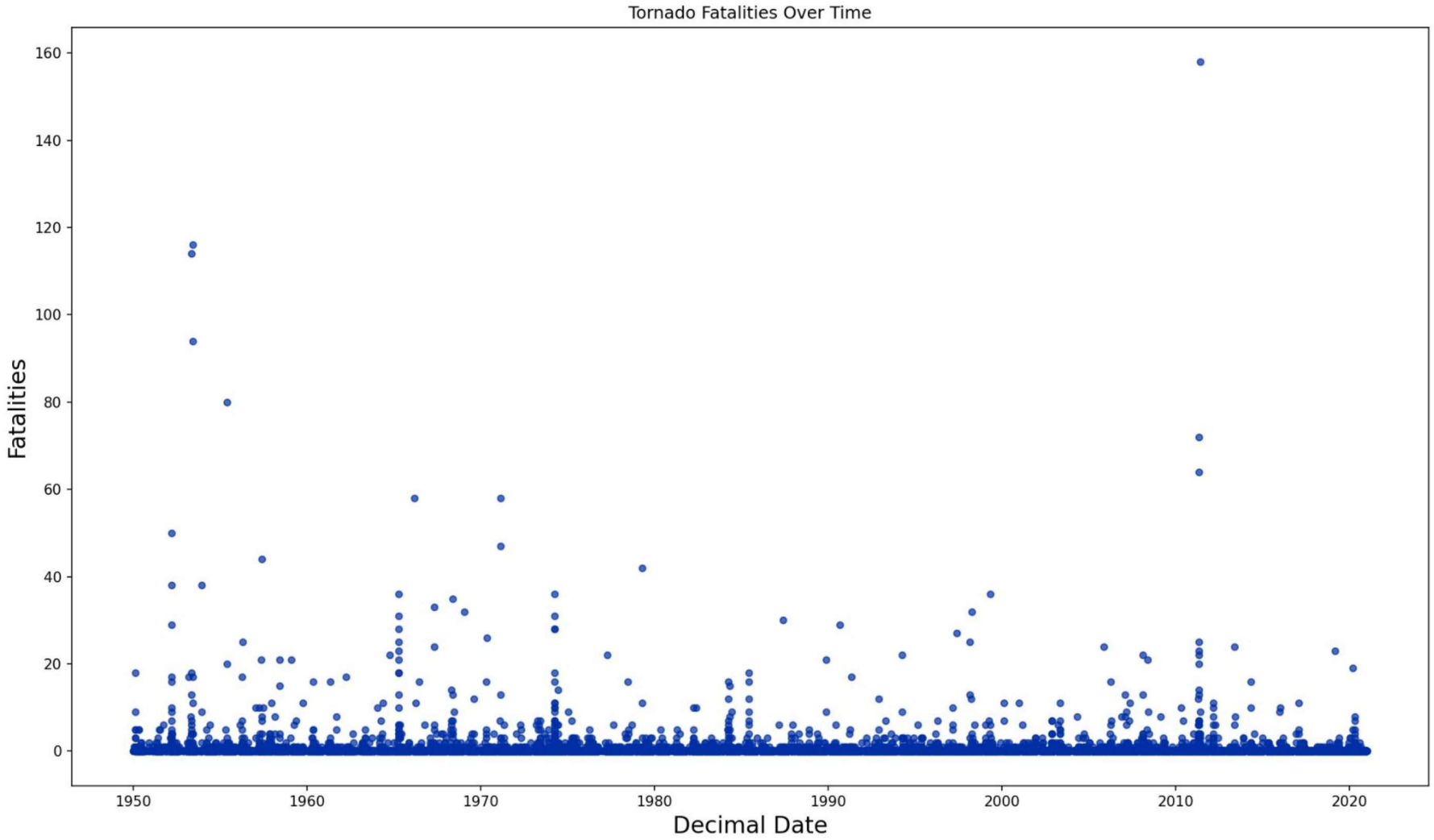

Number of Tornadoes by Year
Count
Year
200
400
600
800
1000
1200
1400
1600
1800
1950
1960
1970
1980
1990
2000
2010
2020

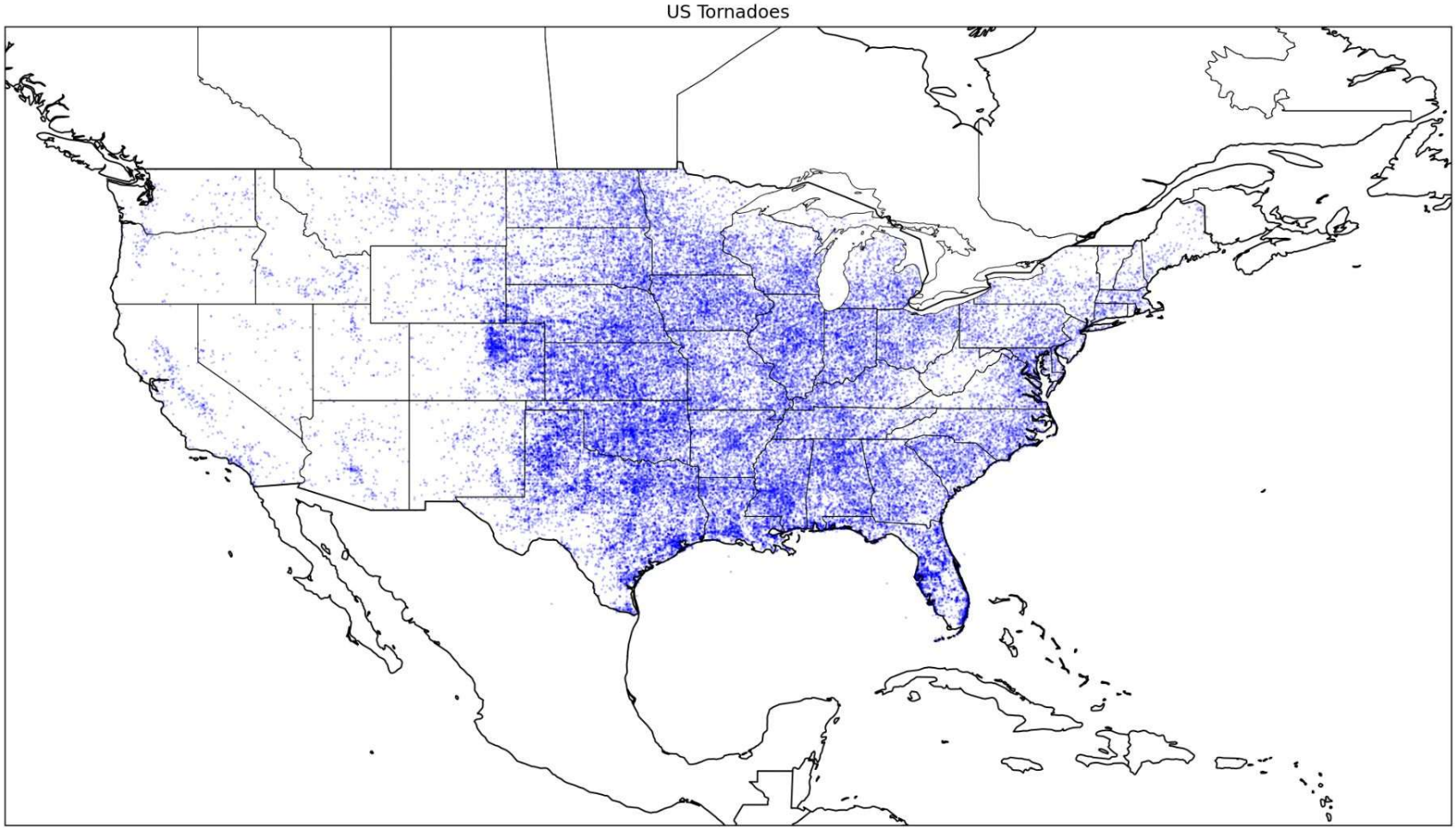

Top 10 Years with the Most Tornadoes
Year
2004
2011
2008
1998
2003
2017
1999
2019
1992
2010
0
250
500
750
1000
1250
1500
1750
Count

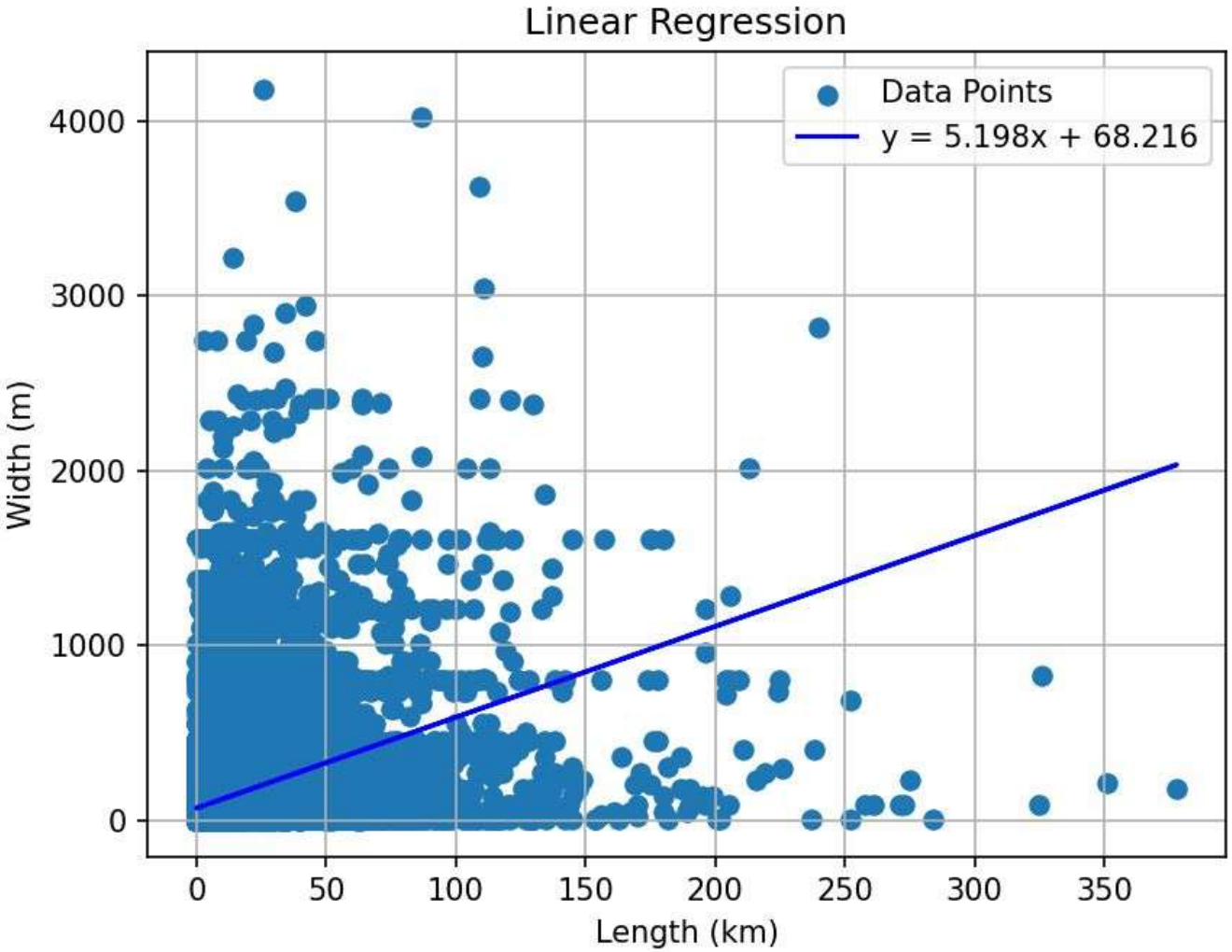

Top 10 Years with the Most EF-5 Tornadoes
Year
1974
2011
1953
1968
1957
1966
1976
1990
1955
1964
0
1
2
3
4
5
6
7
Count

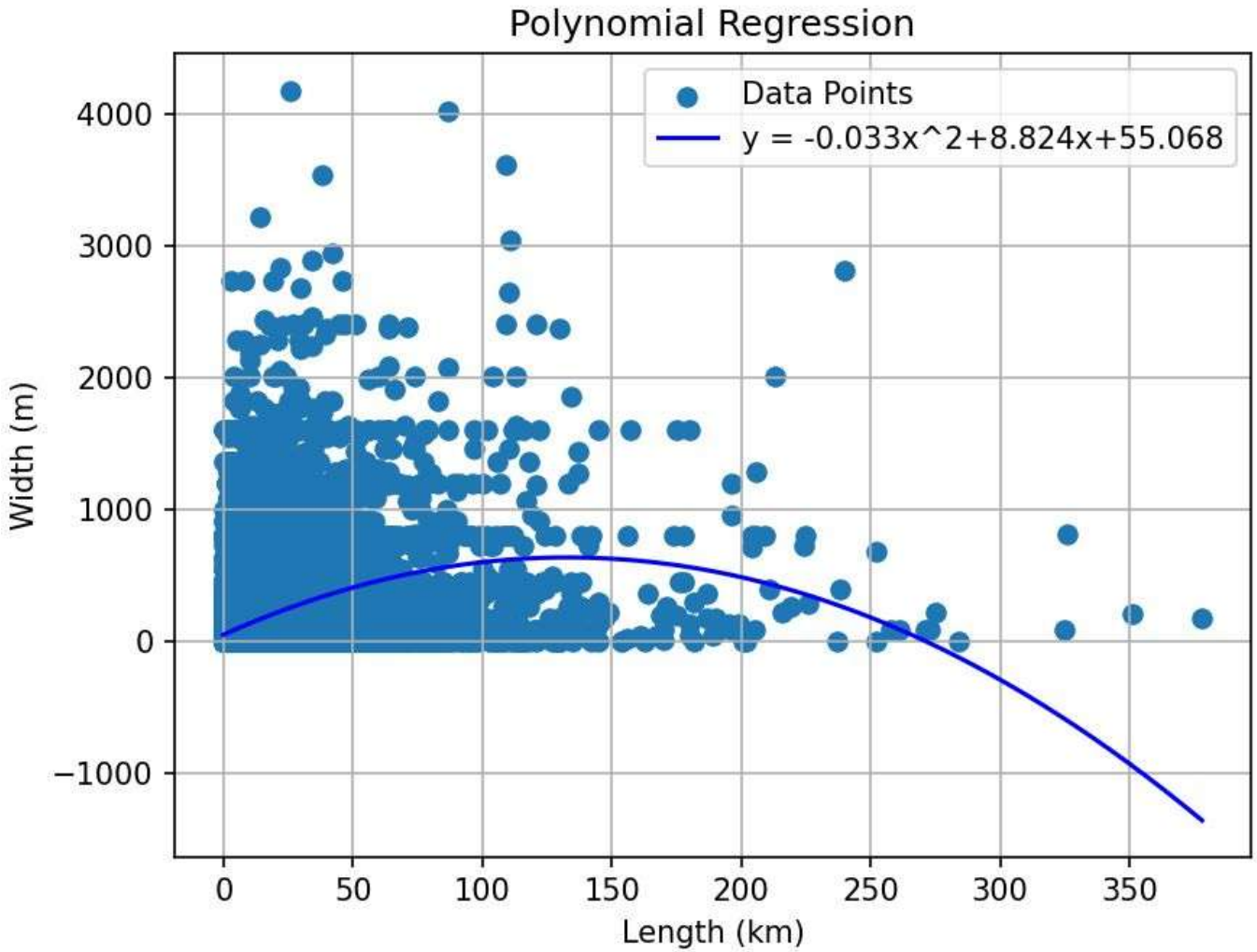

Top 10 Days with the Most Tornadoes
Date
2011-04-27
1974-04-03
2003-06-24
2011-05-25
2007-05-05
1999-01-21
2012-04-14
2003-05-04
2004-05-30
1995-05-18
0
25
50
75
100
125
150
175
200
Count

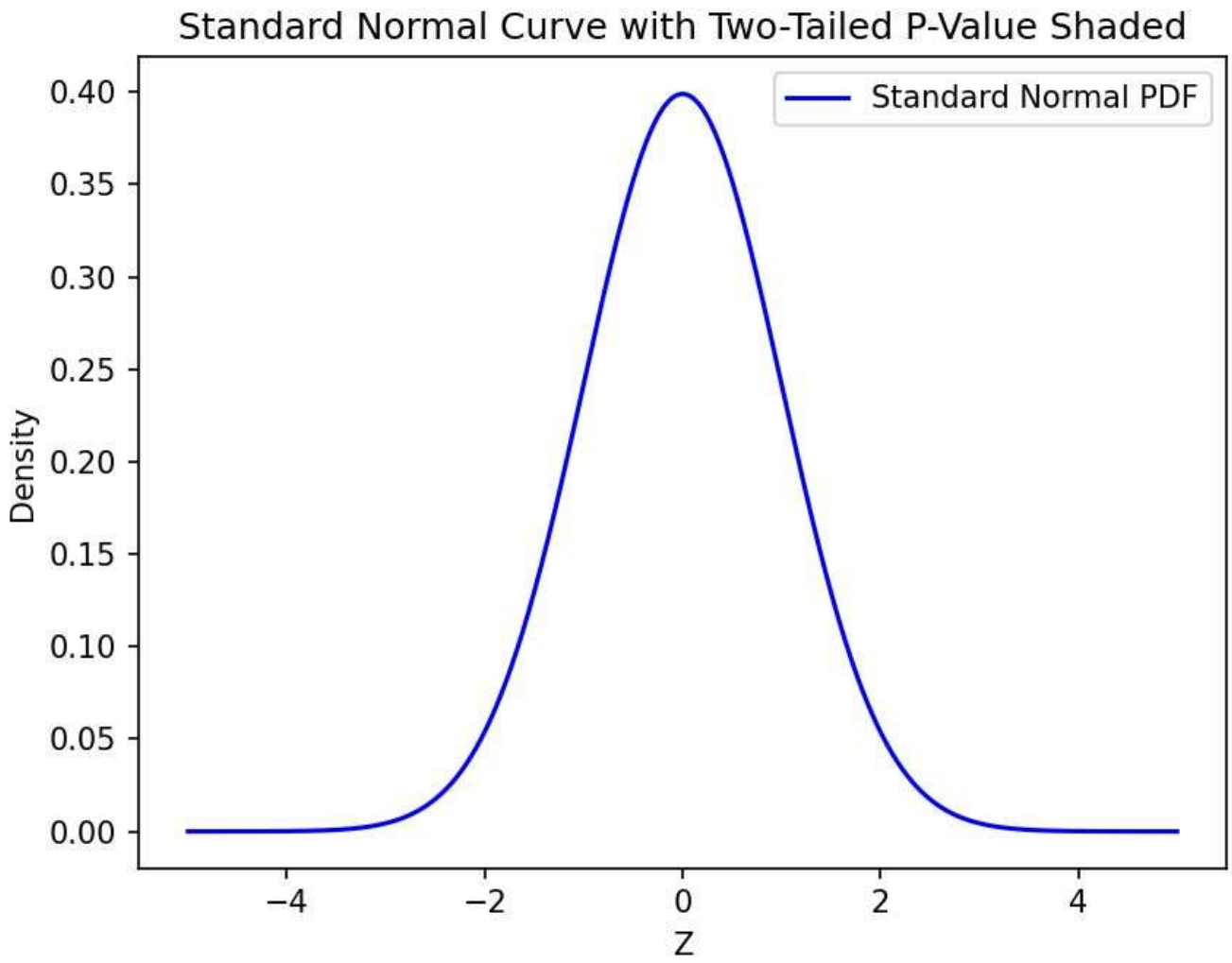

Tornado Fatalities Over Time
Fatalities
Year
0
100
200
300
400
500
1950
1960
1970
1980
1990
2000
2010
2020


Tornado Fatalities Over Time
Fatalities
Decimal Date
0
20
40
60
80
100
120
140
160
1950
1960
1970
1980
1990
2000
2010
2020

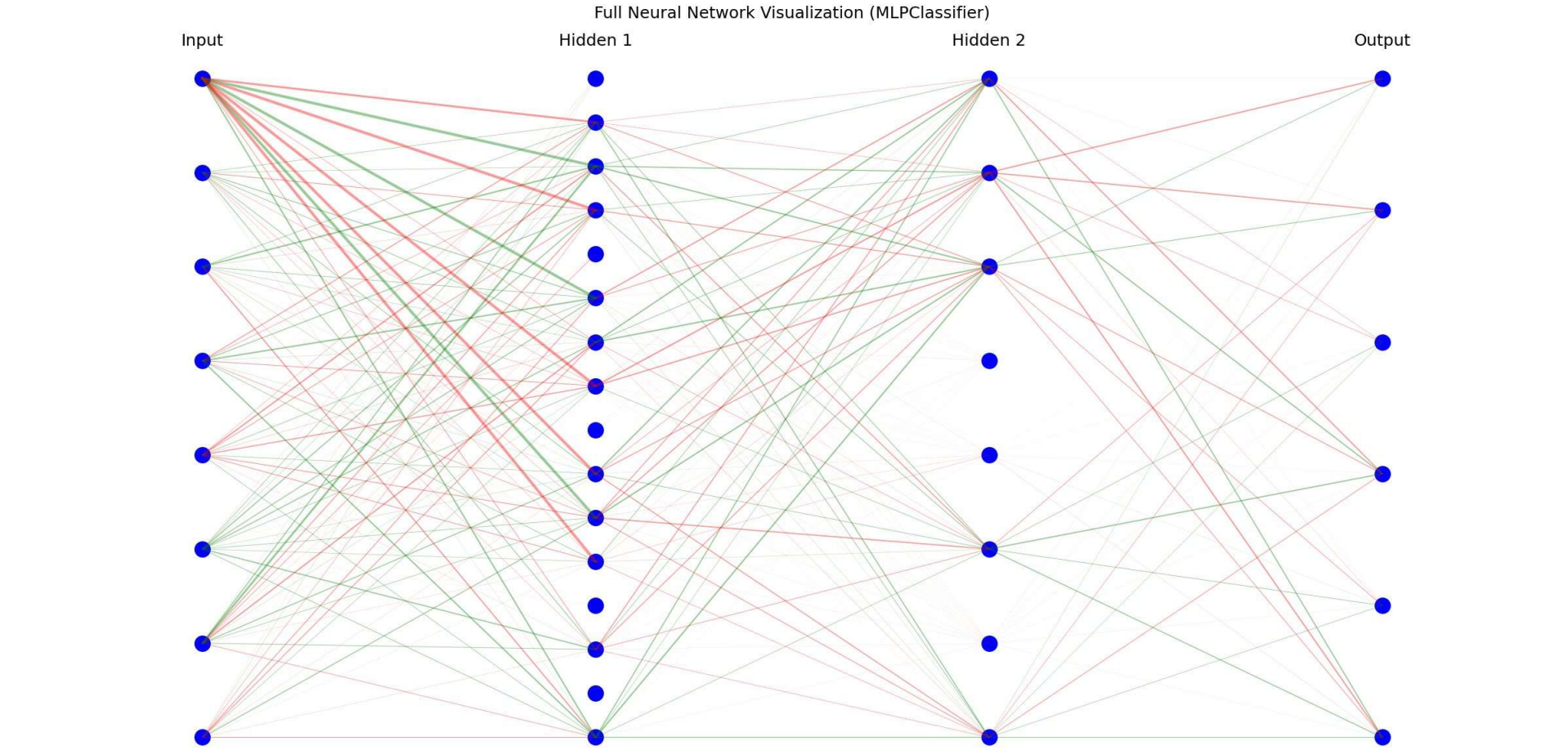

US Tornadoes


Linear Regression
Data Points
y = 5.198x + 68.216
Width (m)
Length (km)
0
1000
2000
3000
4000
0
50
100
150
200
250
300
350

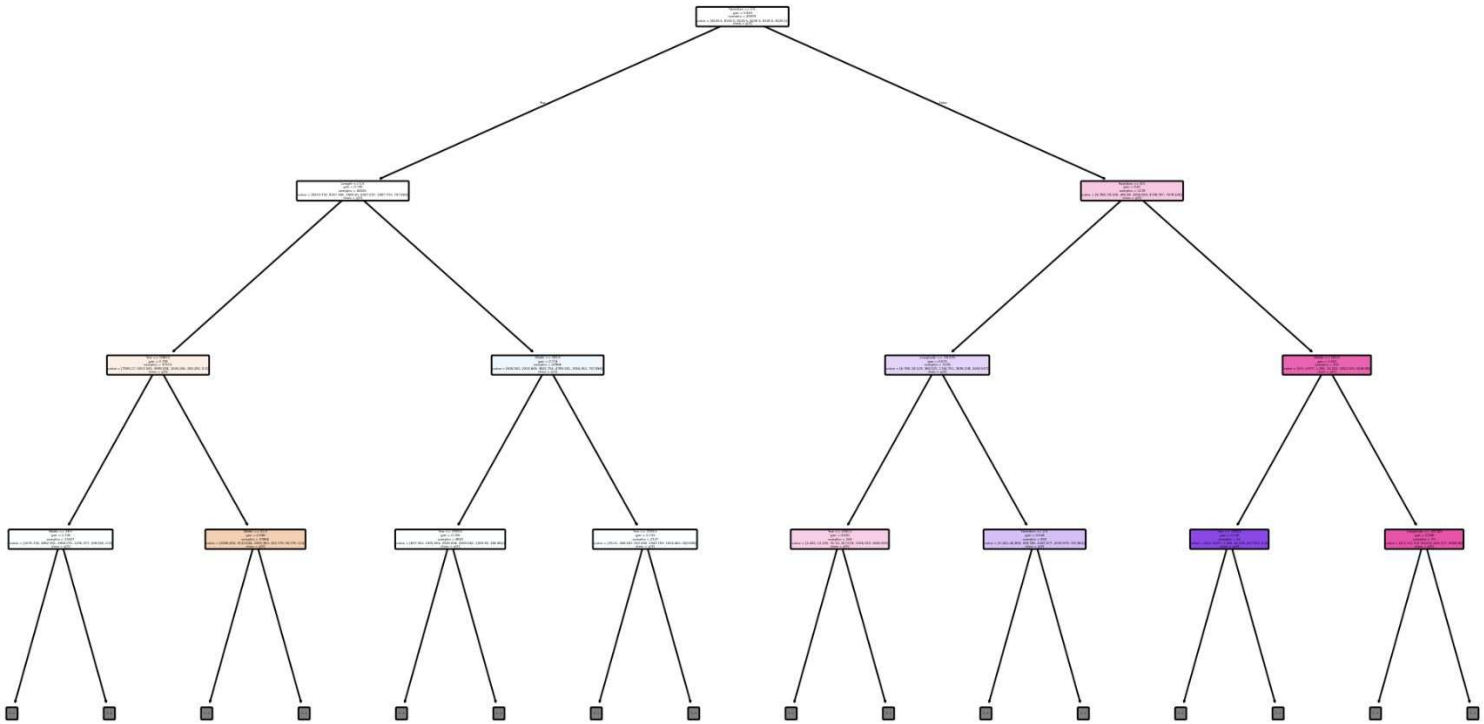

Polynomial Regression
Data Points
y = -0.033x^2+8.824x+55.068
Width (m)
Length (km)
4000
3000
2000
1000
0
−1000
0
50
100
150
200
250
300
350


Standard Normal Curve with Two-Tailed P-Value Shaded
Standard Normal PDF
Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−2
0
2
4
Z

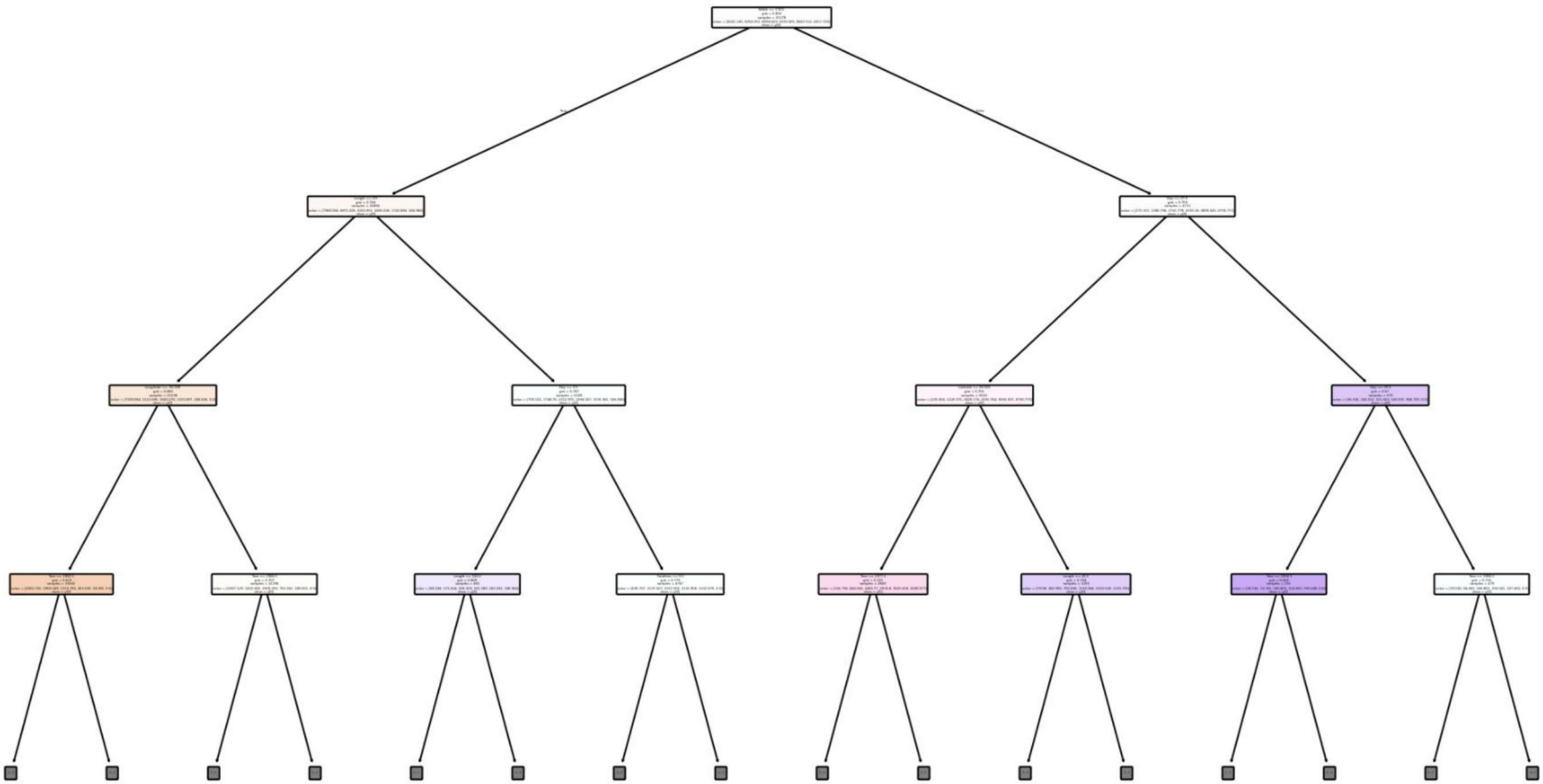
```
Summary Statistics:
Pre-2000s
Mean: 0.9243759721502111
Standard Deviation: 0.9432632399829439
Count: 40503
Post-2000s
Mean: 0.5528644804425128
Standard Deviation: 0.7719593702182752
Count: 25310
Z-Score: 55.06908948080997
P-value (Riemann sum): 0.0
Conclusion: Reject the null hypothesis — there is a significant difference in means.
```



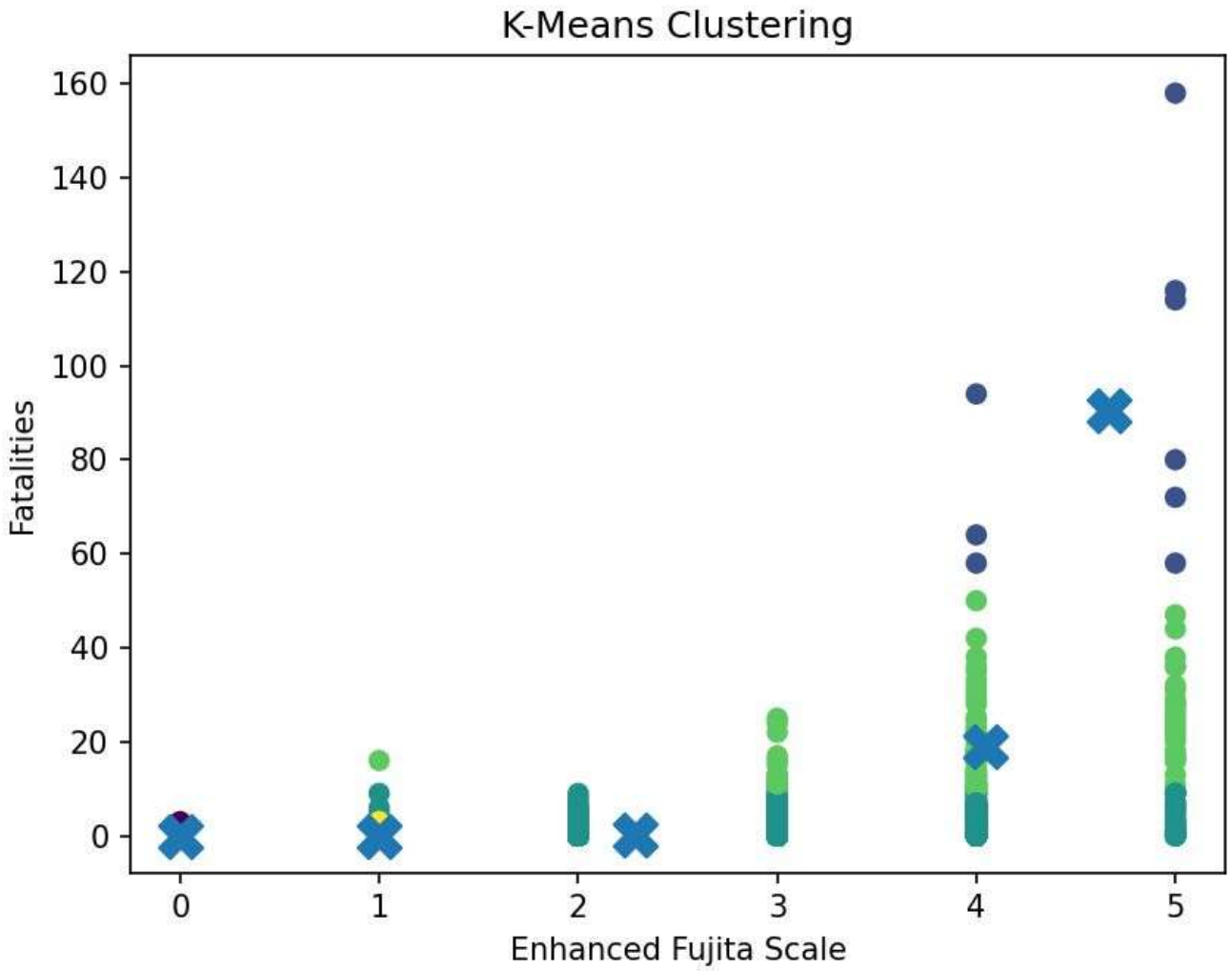

```
Neural Network Prediction (Dallas, TX @ May 25, 2025 | 10 km traveled, 500 m wide, 25 deaths): EF-5
Neural Network Accuracy: 0.5723
```



```
Decision Tree Prediction (Dallas, TX @ May 25, 2025 | 10 km traveled, 500 m wide, 25 deaths): EF-4
Decision Tree Accuracy: 0.5568
```



```
Random Forest Prediction (Dallas, TX @ May 25, 2025 | 10 km traveled, 500 m wide, 25 deaths): EF-4
Random Forest Accuracy: 0.6577
```


K-Means Clustering
Fatalities
Enhanced Fujita Scale
0
20
40
60
80
100
120
140
160
0
1
2
3
4
5